

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think> To solve this , we first need to find the momentum of the electron using its kinetic energy . The kinetic energy (KE) of an electron in terms of energy and momentum is given by $(KE = \frac { 1 }{ 2 } mv ^ 2 \)$, but for non-relativistic speeds (which is the case for electrons with 1 e V energy), we can also use $(KE = p ^ 2 / 2 m \)$, where $(p \)$ is the momentum of the electron and $(m \)$ is the mass of the electron .

Given , $(KE = 1 \text{ eV} \)$, we can convert it to joules : $(1 \text{ eV} = 1 . 6 \times 10 ^ { - 19 } \)$ joules .

Now , using the formula $(KE = p ^ 2 / 2 m \)$, where $(m \)$ (mass of the electron) is $(9 . 11 \times 10 ^ { - 31 } \)$ kg , we solve for $(p \)$:

$$(1 . 6 \times 10 ^ { - 19 } = p ^ 2 / 2 (9 . 11 \times 10 ^ { - 31 }))$$

$$(p ^ 2 = 2 . 92 \times 10 ^ { - 19 } \times 2 (9 . 11 \times 10 ^ { - 31 }))$$

$$(p ^ 2 = 5 . 398 \times 10 ^ { - 30 })$$

$$(p = \sqrt { 5 . 398 \times 10 ^ { - 30 } } = 7 . 34 \times 10 ^ { - 15 }) \text{ kg m/s} .$$

The de Broglie wavelength (λ) is given by $(\lambda = h / p)$, where (h) is Planck's constant $(6 . 626 \times 10 ^ { - 34 } \text{ Js})$.

$$(\lambda = (6 . 626 \times 10 ^ { - 34 }) / (7 . 34 \times 10 ^ { - 15 }))$$

$$(\lambda = 8 . 99 \times 10 ^ { - 10 }) \text{ m or } (8 . 99 \times 10 ^ { - 8 }) \text{ cm} .$$

Among the given choices , $(8 . 66 \times 10 ^ { - 6 })$ cm is close . But it's generally specified the wavelength is around $(\lambda = 10 ^ { - 8 } \text{ cm})$, which can be found from options given .

Therefore , the most suitable from the choices given was $(8 . 66 \times 10 ^ { - 6 } \text{ cm})$.

</think>
<answer> $8 . 66 \times 10 ^ { - 6 } \text{ cm}$ </answer>